



SIMATS
SCHOOL OF ENGINEERING
Approved by AICTE | IET-UK Accreditation



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Compiler-Based Disaster Relief Resource Allocation for Faster and Accurate Emergency Response

CSA1407-COMPILER DESIGN

Presented by:

Rekha S K (192511023)

Pallavi P (192525448)

UNDER THE GUIDANCE

Dr. N. Deepa



INTRODUCTION



- Natural disasters such as floods, earthquakes, cyclones, and fires require quick and efficient resource allocation.
- Traditional disaster management uses manual verification, causing delays.
- This project applies compiler design concepts like lexical analysis, syntax checking, semantic validation, and rule-based execution to process emergency requests.
- The system validates inputs, allocates resources, and generates reports automatically, improving response speed, reducing human errors, and ensuring accurate relief resource allocation.



Website Deployment & Software Requirements



Website Deployment

- Frontend: HTML, CSS
- Backend: C Programming
- Database: MySQL
- IDE: Visual Studio Code
- Compiler: GCC Compiler
- Deployment: Localhost

Software Requirements

- Visual Studio Code
- GCC Compiler
- HTML
- CSS
- MySQL



Existing Limitations



- Existing disaster management systems depend on manual verification, increasing response time during emergencies.
- Most systems lack compiler-like syntax and rule validation before processing disaster requests.
- Resource allocation is often manual, causing inefficient distribution of relief materials.
- They also have limited automation in report generation, resource tracking, and prioritizing emergency requests based on disaster severity and available resources.



Module 1 – Input Processing and Rule Validation



- User enters disaster details.
- System checks valid input format.
- Invalid inputs are rejected.
- Logical errors are identified.
- Valid requests move to the allocation module.

Compiler Phases Used

1. Lexical Analysis
 - Identifies valid words and symbols.
2. Syntax Analysis
 - Checks whether the input follows the correct format.
3. Semantic Analysis
 - Verifies logical correctness.



Module 2 – Resource Allocation Rule Engine



- Reads validated requests.
- Applies predefined allocation rules.
- Assigns food, water, medicine, and rescue teams.
- Prioritizes high-risk disaster areas.
- Stores allocation details in the database.

Compiler Phases Used

4. Intermediate Code Generation

- Converts validated input into internal instructions.

5. Code Optimization

- Removes unnecessary processing.
- Allocates resources efficiently.

